

CHAHAT KUMAR

Delhi, India

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Generative AI. Experienced in developing end-to-end AI applications involving data preprocessing, model training, evaluation, explainability, real-time inference, and deployment. Strong technical experience with Python, PyTorch, OpenCV, ConvNeXtV2, MediaPipe, ONNX Runtime, Scikit-learn, Streamlit, and Generative AI APIs. First author of research work focused on imbalance-aware and explainable facial emotion recognition. Experienced in developing practical AI solutions for computer vision, recommendation systems, NLP, predictive analytics, and intelligent automation.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Deep Learning: PyTorch, TensorFlow, Keras, Artificial Neural Networks, Convolutional Neural Networks, Transfer Learning, ConvNeXtV2

Computer Vision: OpenCV, MediaPipe, YOLO, Face Detection, Facial Emotion Recognition, Image Classification, Pose Detection, Image Preprocessing

Generative AI: Gemini API, Groq API, Prompt Engineering, Generative AI Application Development

NLP: Text Preprocessing, Tokenization, Feature Extraction, Text Classification

Model Deployment: ONNX Runtime, Streamlit, Real-Time Inference

Data and Analytics: NumPy, Pandas, Matplotlib, Seaborn, Data Visualization, Dashboard Development

Tools and Platforms: Git, GitHub, Jupyter Notebook, VS Code, Kaggle

PROFESSIONAL EXPERIENCE

Project Assistant / Teaching Support – Part-Time

NIELIT, Delhi

- Assisted with Artificial Intelligence, Python, Machine Learning, and IT-related academic projects.
- Guided learners in Python programming and fundamental Machine Learning concepts.
- Supported practical laboratory sessions, project implementation, debugging, and technical problem-solving.
- Assisted learners with the practical application of Artificial Intelligence and programming concepts.
- Provided technical support during project development and training activities.

RESEARCH EXPERIENCE

First Author – Explainable Imbalance-Aware Facial Emotion Recognition

NIELIT, Delhi

- Developed an imbalance-aware facial emotion recognition framework using the ConvNeXtV2-Tiny deep learning architecture.
- Applied transfer learning and self-supervised FCMAE pretraining to improve feature representation and generalization.
- Addressed severe class imbalance using Class-Weighted Focal Loss.
- Achieved a macro-F1 score of **0.94** across emotion classes.
- Achieved strong minority-class recognition, including a **0.91 F1-score for the Fear class**.
- Integrated Grad-CAM for model interpretability and validation of semantically meaningful facial regions.
- Evaluated performance against CNN-based benchmark architectures including DenseNet121 and EfficientNetV2.
- Manuscript in preparation.

SELECTED AI/ML PROJECTS

Real-Time Mood-Based Song Recommendation System

PyTorch / ConvNeXtV2 / OpenCV / ONNX

Runtime / Spotify API

- Developed a real-time facial emotion recognition pipeline using ConvNeXtV2.
- Achieved over **94% emotion classification accuracy**.
- Implemented face detection, image preprocessing, emotion classification, and real-time inference using OpenCV.
- Integrated emotion recognition with mood-based music recommendation functionality.
- Optimized model inference using ONNX Runtime for efficient real-time execution.

AI Fitness Coach

Python / MediaPipe / Streamlit / Gemini AI / Computer Vision

- Developed an AI-powered fitness application combining Computer Vision and Generative AI.
- Implemented body landmark detection using front, side, and back images.
- Developed a body-analysis workflow for evaluating posture and body proportions.

- Integrated Gemini AI to generate personalized workout plans and diet recommendations.
- Designed an interactive Streamlit interface with analytics and progress tracking.

AI Interview Coach

JavaScript / Generative AI / Web Application

- Developed an AI-powered interview preparation application.
- Implemented an interactive workflow for AI-assisted interview practice and feedback.
- Applied web technologies and Generative AI concepts to build a practical career-development application.

YouTube URL-to-Script Generator

Python / Generative AI / AI Application Development

- Developed an AI application for processing YouTube video URLs and generating structured scripts.
- Applied AI-based content processing and automated text generation.
- Built the application as a practical Generative AI and content automation project.

Laptop Price Prediction System

Python / Machine Learning / Regression

- Developed a machine learning system for laptop price prediction.
- Performed data preprocessing, feature engineering, model training, and evaluation.
- Developed a prediction interface for real-time price estimation.

Book Recommendation System

Python / Pandas / Scikit-learn / Recommendation Systems

- Developed a recommendation system using content-based and popularity-based approaches.
- Implemented similarity-based filtering for personalized recommendations.
- Performed dataset processing and feature engineering to improve recommendation relevance.

SMS Spam Detection System

Python / NLP / Scikit-learn

- Developed a machine learning classifier for SMS spam detection.
- Implemented text cleaning, tokenization, feature extraction, and supervised classification.
- Built an end-to-end NLP pipeline for binary text classification.

Olympics Analytics Dashboard

Python / Pandas / Matplotlib

- Developed a data analytics dashboard for historical Olympic performance analysis.
- Processed structured datasets and generated country-wise performance insights.
- Created visualizations to identify historical trends and performance patterns.

CERTIFICATIONS AND TRAINING

- NIELIT A Level – Advanced Information Technology
- Machine Learning Certification
- Python Programming
- Artificial Intelligence
- Artificial Intelligence Bootcamp
- Cloud Computing
- Internet of Things (IoT)
- Web Development
- Kaggle – Computer Vision
- Kaggle – Introduction to Deep Learning
- Kaggle – Introduction to SQL
- Kaggle – Time Series

NIELIT QUALIFICATIONS

- O Level – Information Technology
- A Level – Advanced Information Technology

ONLINE PORTFOLIO

GitHub: <https://github.com/Chahat2830>

Portfolio: <https://chahat2830.github.io/-Portfolio-website/>

LinkedIn: <https://www.linkedin.com/in/chahat-kumar-34475231a/>

AREAS OF INTEREST

Artificial Intelligence, Machine Learning, Computer Vision, Deep Learning, Generative AI, Explainable AI, Real-Time Vision Systems, Natural Language Processing, Recommendation Systems, Intelligent Automation